# HARSHVEER SINGH

**Senior ML/AI Engineer | Agentic & LLM Systems · Deep Learning · Time-Series · Production ML**

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## SUMMARY

ML engineer with 5+ years taking deep learning from research to production: agentic and multi-agent LLM systems, sequence models over streaming audio and long documents, multilingual transformers, and sub-200 ms edge inference. Currently building agentic AI for capital markets. Hands-on MLOps (model registries, regression suites, CI/CD, cloud and on-device serving). Research grounding in optimization and loss-landscape geometry (CERN CMS, IIT Mandi), now applied to time-series and market data.

## TECHNICAL SKILLS

- **LLM & agentic systems:** multi-agent orchestration, swarm-style agent coordination, tool calling, RAG, vLLM, llama.cpp, LoRA/PEFT, Unsloth, LangChain, LlamaIndex, vector DBs (Weaviate, ChromaDB)
- **Deep learning & sequence models:** PyTorch, TensorFlow, Hugging Face Transformers, CNN-GRU/RNNs, CTC, attention pooling, contrastive & multitask learning, knowledge distillation, quantization (ONNX, TFLite INT8)
- **Time series & statistics:** change-point / structural-break detection (CUSUM, GLR), AR/GARCH, distribution-shift testing, walk-forward and purged/embargoed CV, backtesting, Bayesian inference, linear algebra (spectral methods, SVD/QR), Hessian and loss-landscape analysis
- **Systems & MLOps:** Python, C/C++, SQL, Bash, Linux, async Python (asyncio/aiohttp), Kafka, Redis, PostgreSQL, Docker, Kubernetes, CI/CD, SageMaker, VertexAI, TorchServe, model registries

## PROFESSIONAL EXPERIENCE

**Senior ML/AI Engineer · XLNC (Brexy)** Jul 2026 – Present

*Singapore-based, remote · agentic AI for capital markets*

- Building agentic AI for capital-markets workflows: multi-agent orchestration, tool calling, and swarm-intelligence coordination across specialized agents.
- Building the agent studio, where agents are configured, deployed and collaborate through a shared messaging layer.

**Senior ML/AI Engineer (Research) · ToneTag Pvt. Ltd.** Jul 2025 – Jul 2026

*Voice commerce and payments — edge + cloud ML*

- Shipped sub-10MB multitask SLU sequence models (CNN-GRU / Transformer encoders, supervised contrastive loss): **>95% intent/slot accuracy at <200 ms** on ARM CPUs.
- Built a <1MB streaming wake-word detector distilled from a wav2vec2-XLSR teacher: **97% recall at <1% false acceptance** on noisy field audio.
- Led on-device ASR + multitask SLU for a TATA car-command model: Digit-CTC, open-world entity and contact-slot heads, joint multi-loss training on a shared latent space.
- Architected a production multi-agent runtime (supervisor router, domain agents, 16-tool executor, RAG, pluggable LLMs; prompts, tool schemas and flows configurable per vertical) behind VoicePay, SoundBox and Blue Tokai ordering for TATA EV.
- Owned deployment: custom model registry, PyTorch to ONNX/TFLite INT8 with C inference hooks for Android/Linux, async (aiohttp/asyncio) serving replacing thread pools; ML yes/no classifier (96–97%) replacing regex checks.

**Applied AI / Data Scientist · Vaultedge Pvt. Ltd.** Oct 2022 – Jun 2025

*Document AI for US and Indian NBFCs*

- Raised production document-AI metrics **~11% with zero regressions** for US NBFC customers via multilingual transformer classifiers and per-customer evaluation loops.
- Built long-context RoBERTa-based hierarchical classifiers; trained and deployed for HDFC, Hero FinCorp and other NBFCs.
- Designed latent-space losses to separate confusable classes, backed by t-SNE, embedding and confusion diagnostics.
- Long-document clustering and learned segment scoring for needle-in-haystack retrieval, cutting retraining at customer onboarding.
- Deployed LoRA/PEFT-tuned Gemma agentic workflows for ingestion, extraction and validation; CPU-only LLM inference at **~75 tokens/s, 20–30 concurrent users**.
- Owned MLOps: in-house model registry (W&B to S3 versioning), SageMaker regression-test suites, CI/CD to TorchServe endpoints, autoscaled Docker serving for GB-scale PDF workloads.

**NLP Engineer, AI Team · Mercer-Mettl** Jan 2021 – Sep 2022

*Intern (Jan–Jun 2021), then full-time*

- **+25% precision** on sentence-cohesion scoring via a latent-space posterior over BERT features; **+16%** NLI cohesion accuracy as an intern.
- Targeted fine-tuning of the production spell-checker: **+13% recall, +32% precision**.

- C++-accelerated LDA topic modelling for email-formality checks; poly-encoder (Blender) interview-screening bot with answer-bank retrieval and voice input.

## QUANTITATIVE RESEARCH & MARKETS (INDEPENDENT)

**ADIA Lab Structural Break Challenge: Real-Time Edition (CrunchDAO)** Sep – Oct 2026

- **Top ~2%** on the final private leaderboard (36th of ~1,690 teams, 65.31% time-stratified AUC). **Poster selected** for the ADIA Lab Symposium 2026, Abu Dhabi.
- Online change-point detection over 10,000 series: after each observation, output P(break has occurred) from causal features with constant-time updates.
- +2.71 points over 44 runs, 2.27 from replacing one 1,728-feature model with separately trained expert heads (AR/GARCH, CUSUM/GLR, copula-rank) blended at fixed weights.
- Showed a single-split estimate overstated the score by 2+ points; gated submissions on paired out-of-fold tests (local gains transferred at 0.8–1.4×). Documented negative results: sequence models, Chronos features, synthetic data, learned stacking.

**ChartChatAI · signal research, backtesting & multimodal chart analysis** 2024 – Present

- Event-driven backtester over equity and commodity universes with transaction-cost, slippage and borrow modelling; walk-forward validation with purged/embargoed CV and per-regime decomposition.
- Cross-sectional signal pipeline (OHLCV, volume profile, microstructure): ranking, neutralization, turnover-constrained blending, correlation-budgeted selection; Sharpe/drawdown/capacity attribution.
- Fine-tuned quantized Llama 11B-Vision + RAG for chart analysis and hypothesis generation; agentic flow to live paper trading on a broker API. Launched on Product Hunt.

**WorldQuant BRAIN** 2026 – Present

- Cross-sectional equity alpha research on the BRAIN simulation platform.

## RESEARCH

**Quasi-Linear Attention for Rare-Event Discrimination · CMS Experiment, CERN** Jan – May 2021

- Attention architecture separating rare collision events from background: an extreme low-SNR problem over high-dimensional detector data.

**Batch Normalization and Loss-Landscape Geometry · IIT Mandi, Dept. of Mathematics** Dec 2019 – Jan 2020

- Hessian spectral analysis of how batch normalization smooths the loss surface and improves conditioning.

**Independent research** 2018 – 2020

- Beta-2 variation schedule for Adam (2018; related result later in Reddi et al.); cross-layer residual transformer for smoother loss surfaces (2020); adversarial self-play for dialogue agents, one agent discriminating the other against human dialogue (2020); poly-encoder fine-tuning of GPT-2 (2020).
- Repodox: RAG over public and private GitHub repositories for code-level Q&A (Weaviate, llama.cpp).

## EDUCATION & MORE

**B.E., Electronics & Computer Engg. · Thapar Institute of Engg. & Tech.** 2017 – 2021

- Talks: Causality in Variational Inference and EM (TIET, 2020); Inductive Bias in ML Models (TIET, 2019); Constraining the Posterior to Gaussian in VAEs (IIT Bombay Techfest, 2017).
- Hackathons: SATURNALIA 2017 (1st), PEC-FEST 2017 (2nd), IIT Bombay Techfest 2018 (3rd).